

REAL-TIME OPERATING SYSTEMS

It's a Dynamic World in There

The world of embedded systems is growing fast albeit without much ado. Almost every well-advertised development in the computing space, ranging from 64-bit computing to multi-core processors, has been reflected in the embedded systems space as well. As a result, embedded real-time operating systems have also grown correspondingly to enable developers and manufacturers to make the most of key developments in the hardware space



Robots assembling vehicle floors—a fine example of embedded real-time operating systems

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Gadgets and machines are becoming increasingly powerful. This means that the computing systems embedded in them are also becoming more potent and smarter. Earlier, we used to associate multi-core and such concepts with supercomputers, but then they invaded the personal computing space, and now they are common in embedded systems as well. These changes show that how embed-

ded systems are getting stronger today. Thus, as a complementary product, real-time operating systems (RTOS) are also becoming more powerful.

Let us take a quick look at the recent trends in the RTOS space—in terms of technology, what the industry majors are offering, and the changing requirements of the market.

Faster time-to-market

“New market scenarios indicate that the time-to-market should be the deciding factor in picking up a real-time operating system (RTOS).

Although in the past, RTOS’ selection was determined by an assessment of the functional characteristics of the candidate system, the increased complexity and the changes in the embedded market means that these factors are no longer applicable. With the emphasis on seizing windows of market opportunity, factors which affect time-to-market have become much more important,” says Sunil Kumar Ross, country manager, Ammos Software Technologies Pvt Ltd, speaking for Express Logic, a leading player in the RTOS space.

RTOS must also be capable of meeting the functional requirements of the application. But, Ross notes that using an RTOS that is “overqualified” for an application may have a negative impact on time-to-market. The reason: the additional capabilities make the RTOS more complicated to use, thus pushing the learning curve for developers higher.

So, the bottom line according to him is to select an RTOS that suits project requirements.

He cites the interrupt response time requirements as an example. The common perception is that the fastest RTOS is the best. But some applications do not require such speed. For example, in a desktop system, the keyboard response of about 10 milliseconds is sufficient to keep up with human typing speed. A faster response provides no additional advantage, as the person doing the typing won't be